CSE2525 · DATA MINING · 2026–27

# Introduction to Data Mining

*What data mining is, how this course is run, and why you should tread with caution — this year with an information-access focus.*

A. Anand (head lecturer) · with J. Urbano & N. Tömen · lecture 1 study deck

PART I

# There are no correct answers — only correct design decisions

*Data mining is a craft of choices under messy data. The next two stories — a cholera map in 1854 and an ancient library — are really design decisions about what structure to look for.*

QUESTION

# London, 1854: hundreds are dying of cholera and no one knows why. All you have is a list of the **addresses of the dead**. How would you find the cause?

*No germ theory yet, no microscope data — just locations on a map.*

Think — then → for the answer.

LONDON, 1854

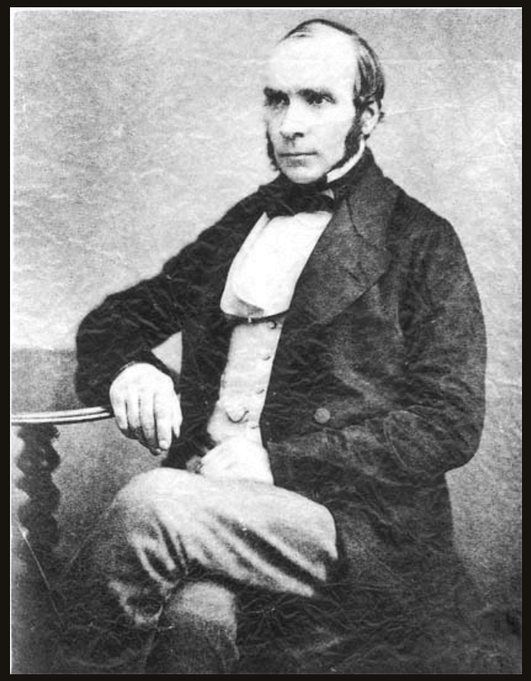
# The first data miner — John Snow

THE PROBLEM

A cholera outbreak tore through Soho — **~600 people dead in days**. The accepted theory was *"bad air"* (miasma). Nobody knew the real cause.

— Snow, a physician, suspected the water — but had no proof.

— All he had was a list of addresses of the dead. How do you find a cause in that?

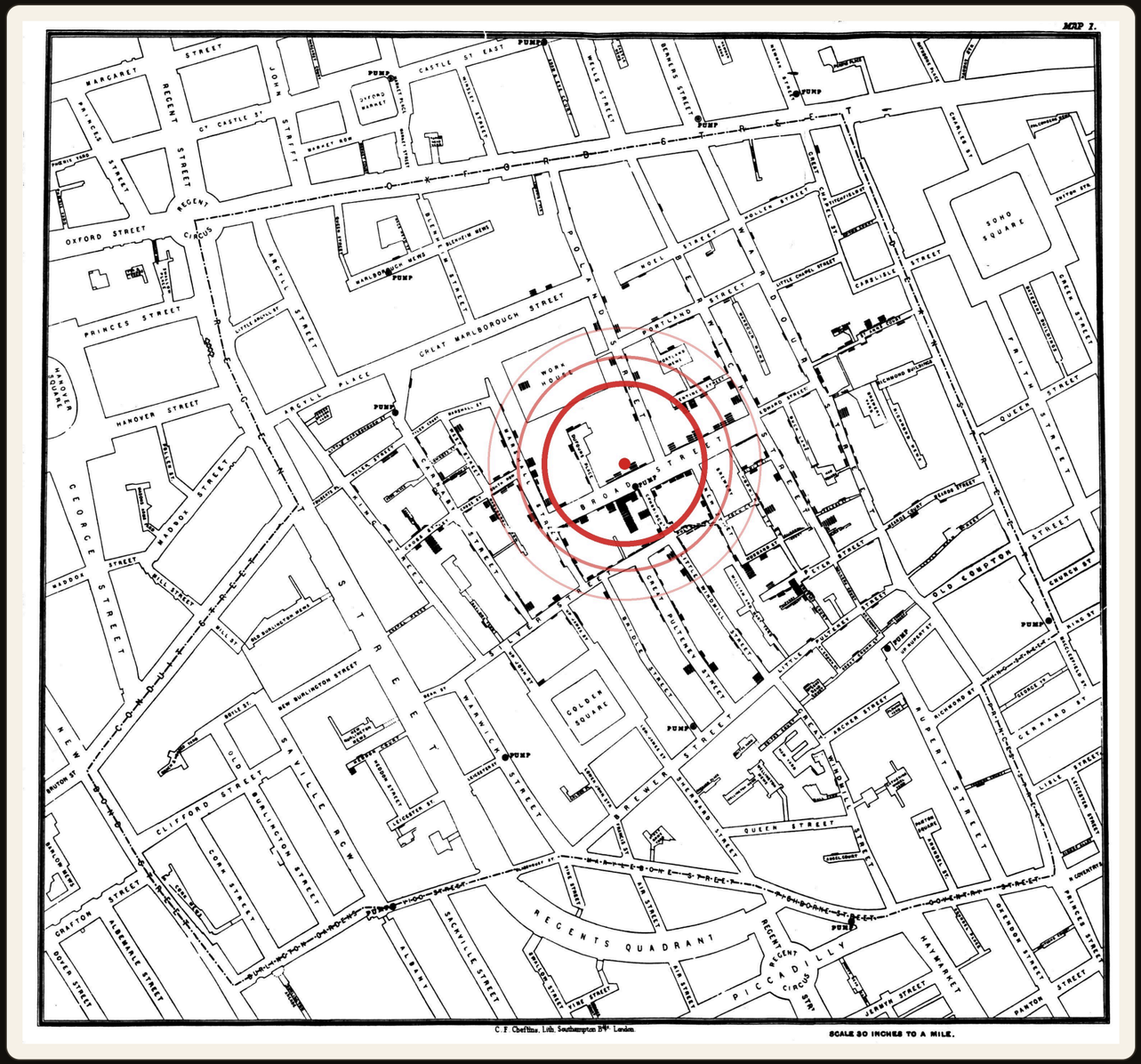


## HE PLOTTED EVERY DEATH ON A MAP




Each red dot is a cholera death. Snow marked them street by street — and a dense cluster began to emerge in the middle of Soho, while whole blocks nearby stayed clear.

# ONE PUMP EXPLAINED IT ALL




The deaths ringed one water pump on Broad Street. Snow had its handle removed — and the outbreak stopped. The design decision — plot deaths in *space*, not in a table — made the cause visible.

ALEXANDRIA, 3RD CENTURY BC

# Another origin story — the first retrieval system

~500,000 scrolls
AN UNUSABLE PILE

→

The Pinakes catalogue
GENRE · AUTHOR · TITLE

→

Any scroll, found
INFORMATION RETRIEVAL

HOW THE PINAKES ORGANIZED IT

Callimachus catalogued the Library of Alexandria — the Pinakes (“tables”), arguably the first information-retrieval system. It split ~half a million scrolls into about ten genres (rhetoric, law, epic, history, medicine…); within each, authors were listed alphabetically, each with a short biography; and every work was recorded by its title, opening line, and line count. Any scroll was now locatable — the first real index.

THE COURSE DEFINITION

# What data mining is — in this course

WORKING DEFINITION

Data mining is the **craft of deciding what structure to look for** in **large, messy, unlabelled** data — and **finding it at scale**.

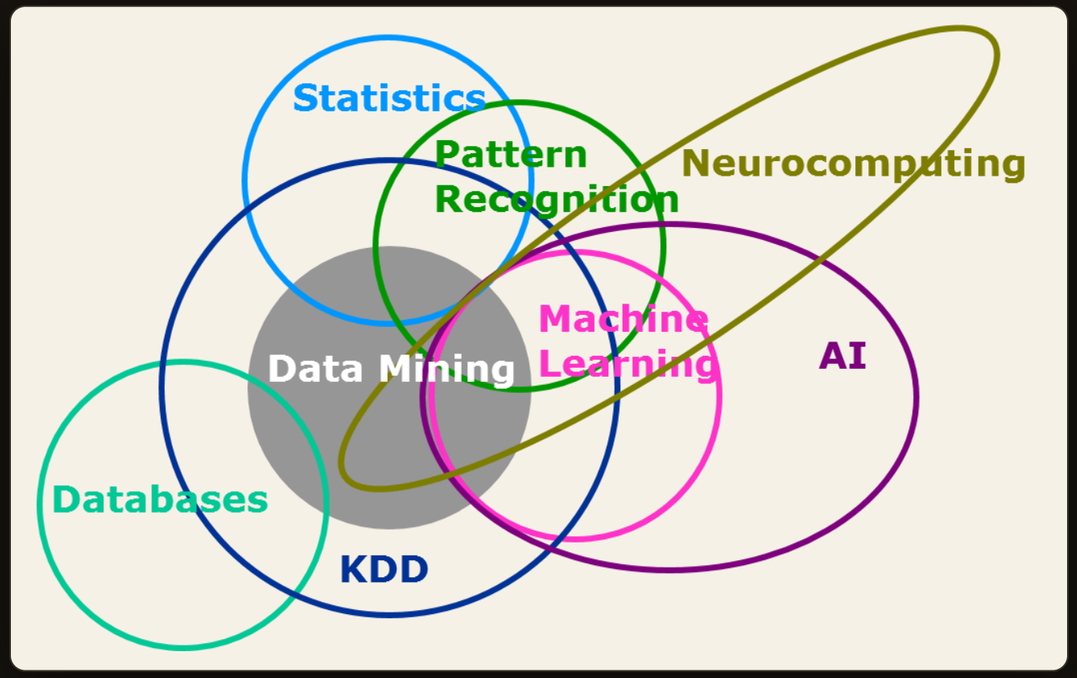
— Deciding what to look for — the design decision, as in Snow's map and the Pinakes.

— Large, messy, unlabelled · at scale — the part of the problem this course focuses on.

A sharper slice of the broader textbook definition ("extract useful information from data"): we emphasise scale and no labels.

BLOGS.SAS.COM

# Data mining sits at an intersection




It borrows from statistics, databases, machine learning, pattern recognition and AI — the overlap at the heart of Knowledge Discovery in Databases (KDD).

THE BROADER TEXTBOOK VIEW

# The textbook definition

CLASSIC DEFINITION

The primary goal of data mining is to **extract useful information from a large volume of data** and transform it into an **understandable structure** for further use.

— “…searching and analyzing a large batch of raw data to identify patterns and extract useful information.”

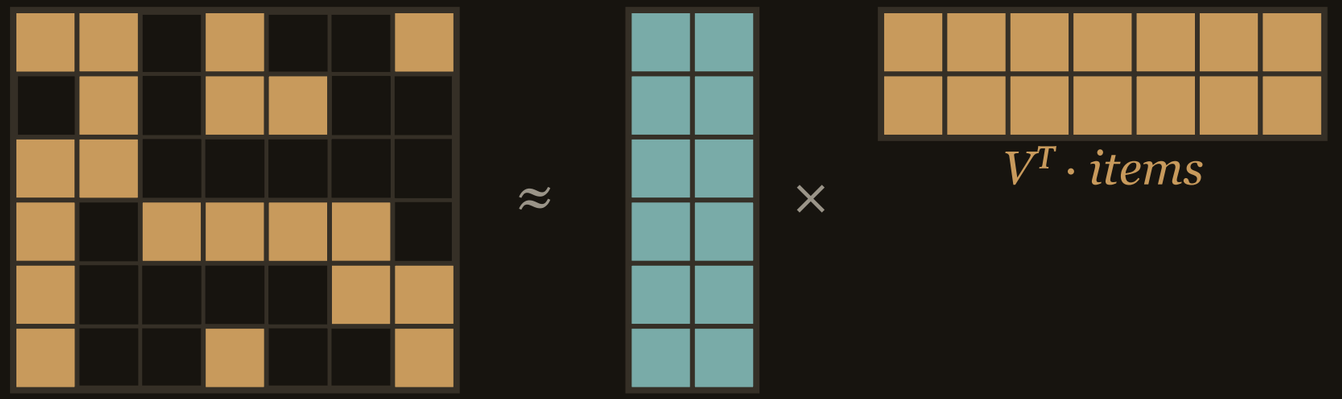
— “…the development of models for data in order to extract information from it.”

— “…analyzing data from different perspectives and summarizing it into useful information.”

— And crucially — it is done by humans.

GUESS FIRST · ① RECOMMENDATION

# How does Spotify know what you'll like?

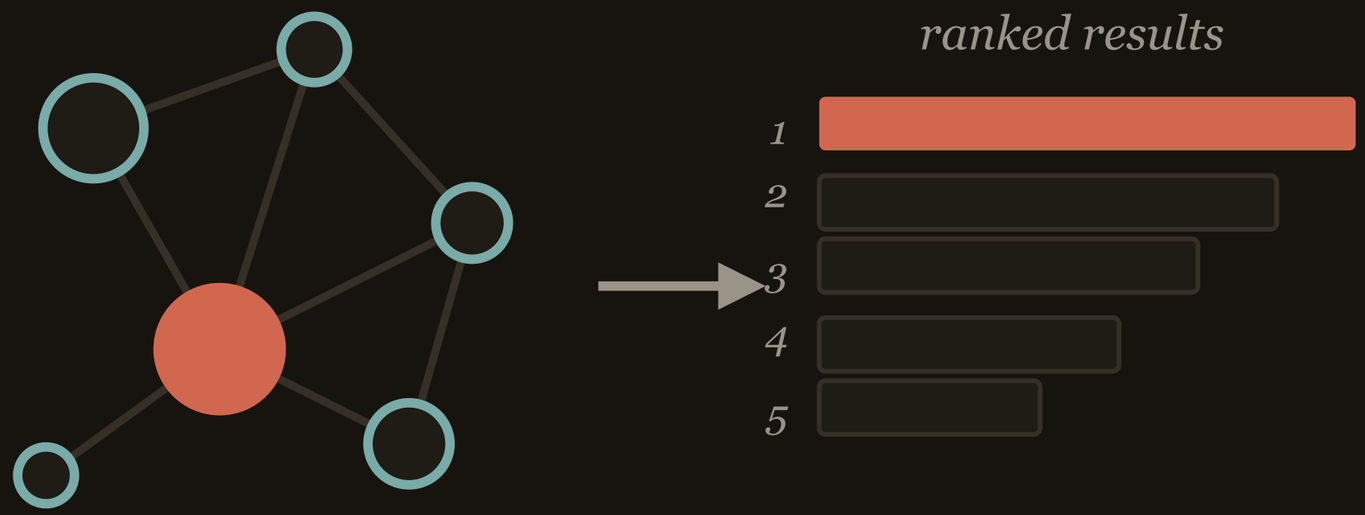



*similar users like similar items*

Distances / similarity | Embeddings | Matrix factorization | Clustering | Dimensionality reduction (PCA)

GUESS FIRST · ② WEB SEARCH

# How does Google rank the entire web for your query?

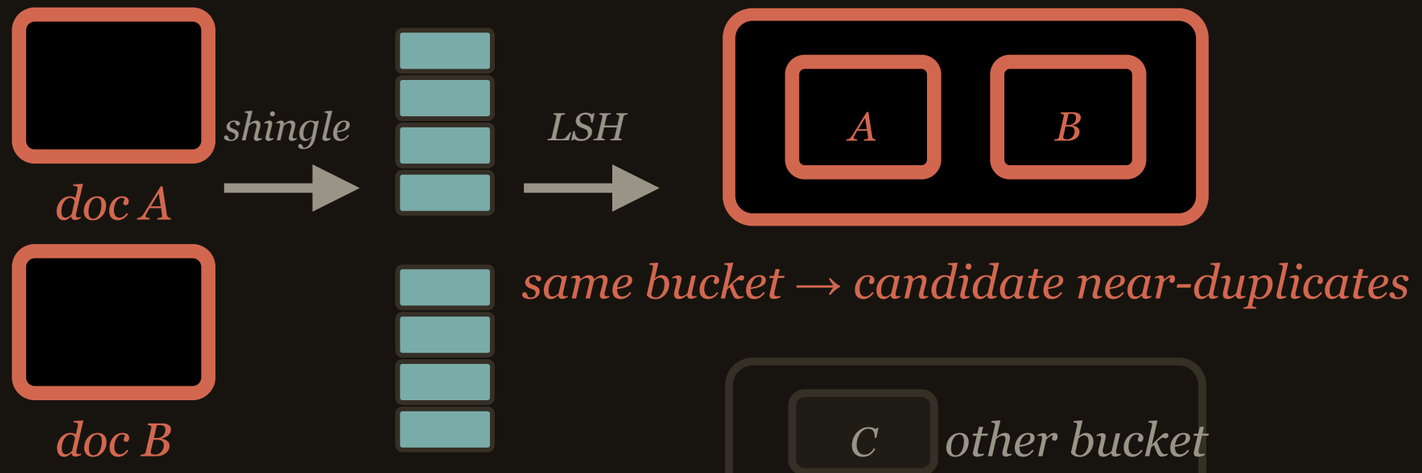



*as a link graph — node size = PageRank*

Inverted index | Graph mining · PageRank | Graph embeddings | Ranking | Retrieval | Evaluation

GUESS FIRST · ③ AT SCALE

# How do you spot near-duplicate pages among billions?




*MinHash signatures*

Jaccard distance | Shingling | MinHashing | LSH / hashing

Fast-filter → slow-check

SAME PIPELINE, THREE DIFFERENT JOBS

# Data mining vs. machine learning vs. data management




Data management — store & serve the data

Data mining — exploration & question-formulation (and making sense of the result)

Machine learning — fitting the model

One shared pipeline — data mining frames the question, machine learning fits the model.

RECAP

# Where data mining lives

THE DIVISION OF LABOUR

Machine learning owns the model-fitting stage — which algorithm, how to train it, the resulting model.

Data mining owns the exploration and question-formulation stage — deciding what to look for and what "similar", "structure", or "relevant" should mean — and then making sense of the result. Often the data is unlabelled, so the learning here is unsupervised.

A HEALTHCARE EXAMPLE

# Mining first, then predicting

DATA MINING PHASE

Analyse a large database of patient records — symptoms, diagnostics, treatments, outcomes — with **clustering**, **association rules**, **pattern discovery** to *uncover patterns* (e.g. fever + cough + shortness of breath → a specific respiratory illness).

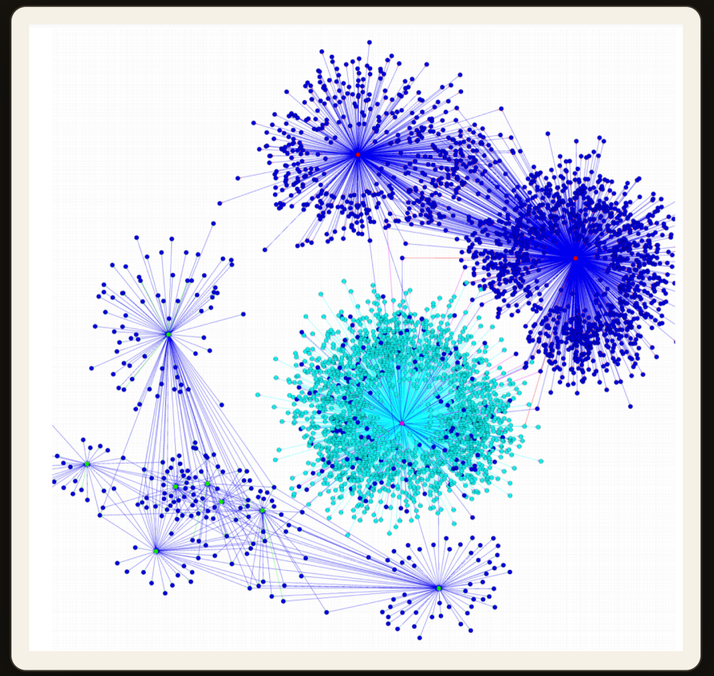
MACHINE LEARNING PHASE

Use those discovered patterns to **train a model**. A new patient's symptoms → the model predicts likely diagnoses and suggests treatments from historical success rates.

Data mining finds the structure; machine learning predicts with it.

CYBER SECURITY · MALWARE FAMILIES

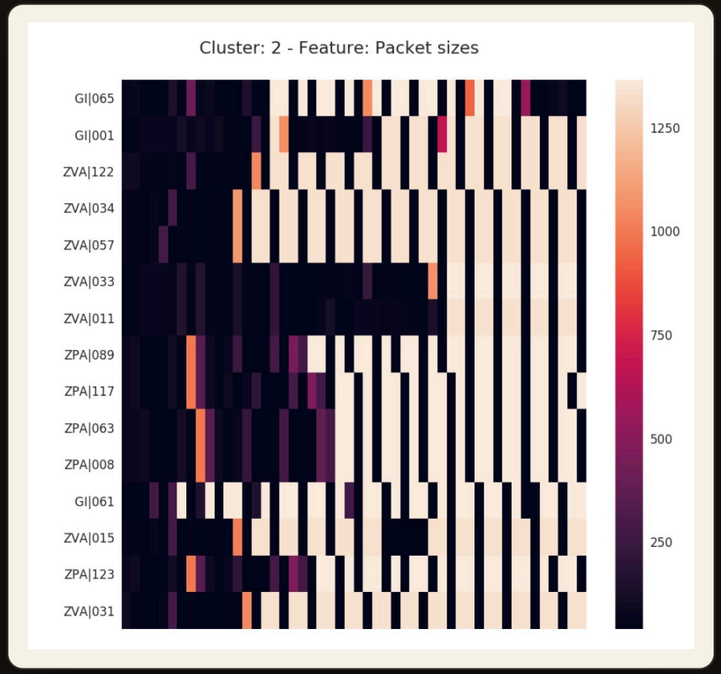
# A real example: network / graph data



Connections between hosts form a huge graph. Clustering it reveals behavioural profiles of malware families — no labels needed (Nadeem et al., 2020).

MAKING THE RESULT INSIGHTFUL

# Visualise groups with heatmaps




Row = connection · column = time · cell colour = bytes transferred. A cluster becomes a readable behavioural signature.

THE WHOLE LOOP

# These are all data-mining skills

THE DATA-MINING LOOP

Take the data as is, but select features using domain knowledge → cluster with clustering methods → make the gained knowledge insightful (e.g. heatmaps).

Feature choice, the right algorithm, and a human-readable result — that end-to-end judgement is the craft.

THE THREE V'S

# Why is data mining hard?

### Volume

- Too big to fit / scan naively
- Need **scalable** algorithms

### Velocity

- Data arrives fast / streams
- Need **approximate yet accurate** estimates

### Variety

- Many types & distributions
- Must **adapt** methods to each

These three pressures are exactly why we need hashing, indexing, embeddings and dimensionality reduction.

PART II

# The course & logistics

*The team, what's new this year, and how everything runs — goals, schedule, labs, grading, prerequisites.*

WHO TEACHES THIS COURSE

# Three instructors, three verticals

## Avishek Anand · head lecturer

- Assoc. prof., Web Information Systems
- IR, NLP, Explainable AI
- Teaches: **distances, embeddings, graphs** — text & graph mining

## Julián Urbano

- Asst. prof., Computer Science (TU Delft)
- Music IR, retrieval evaluation, significance testing
- Teaches: **retrieval, rankings, evaluation**

## Nergis Tömen

- Asst. prof., Intelligent Systems
- Bio-inspired vision, neuromorphic computing
- Teaches: **matrices, PCA, decompositions, recommenders**

WHAT'S DIFFERENT THIS YEAR

# This year: an information-access focus

THE SHIFT

The course is reoriented around information access — how we find, rank, and evaluate the right information inside massive data, not just describe it.

Content swap vs. last year: out go anomaly detection and sketching; in come retrieval, ranking, and evaluation.

— One lab assignment this year — down from two.

— No Kaggle competition this year.

— New reading: Introduction to Information Retrieval (Manning, Raghavan & Schütze) for selected chapters.

WHAT YOU SHOULD WALK AWAY WITH

# Course goals — three strands in every lecture

THE PROMISE

Each lecture runs three strands at once — the **concepts**, the **algorithms**, and the **skills** to apply them.

### Concepts

- The ideas & theory
- Which concept fits which data type

### Algorithms

- Implement them from scratch
- Strengths, weaknesses, and *when* to use them

### Skills

- Pick the right method for a problem
- Make it work on real, messy data

The skills strand lives mainly in the lab project — where you apply concepts & algorithms end-to-end.

THE SPINE OF THE WHOLE COURSE

# Three verticals: Distances · Matrices · Retrieval

### Distances · Avishek

*"When are two things similar — and who decides?"*

- Similarity & metrics, DTW
- Text & graph embeddings
- Clustering · Hashing · MinHash / LSH

### Matrices · Nergis

*"What hidden structure explains the data?"*

- Representation & operations
- Factorization & decompositions
- Dimensionality reduction (PCA)

### Retrieval · Julian

*"How do I find the right thing fast — and how do I know I did?"*

- Indexing & the inverted index
- Ranking · Evaluation
- Near-duplicate / similar-item search

This year's shift: counting folds into Distances, and Retrieval becomes its own vertical — the information-access focus. Every topic hangs off one of these three.

MONDAYS 13:45 · THURSDAYS 10:45 · EEMCS AMPERE · LABS TUE 08:45

# Schedule

| Week | Monday 13:45 | Thursday 10:45 |
|---|---|---|
| 9 Nov | Intro | Distances |
| 16 Nov | Retrieval | Matrices & PCA |
| 23 Nov | Rankings | Evaluation |
| 30 Nov | Embeddings | Matrix decomposition |
| 7 Dec | no lecture | Recommender systems |
| 14 Dec | Hashing | Sketching |
| 21 Dec–1 Jan | ❄ Christmas break — no teaching | |
| 4 Jan | Invited lecture | Indexing |
| 11 Jan | Graphs | Bias & explanations |
| 18 Jan | Exam preparation | no lecture |
| 25 Jan | 📝 **Data Mining EXAM · Mon 25 Jan 13:30** | |

⚑ The first lab is in Week 5 (Tue 8 Dec) — there are no lab sessions until Week 5.

17 LECTURES · 1 LAB

# How the teaching works

- 17 lectures — 13 content, 2 invited, 1 intro, plus Q&A sessions.
- In the room we do case studies — how do you actually apply what you learned to a real problem?
- Complete schedule lives on Brightspace.

WHERE THE CONTENT COMES FROM

# Course material

ON BRIGHTSPACE (REQUIRED)

Lecture slides (after each class), lab exercises (start of each week), and reading materials — selected book chapters and papers.

- Free via the TU Delft digital library: *Mining of Massive Datasets* · *Data Mining*.
- New this year — *Introduction to Information Retrieval* (Manning, Raghavan & Schütze) for the retrieval / ranking / evaluation chapters. Free online: nlp.stanford.edu/IR-book
- Only the selected chapters uploaded to Brightspace are examinable.

MANDATORY · IN STUDENT PAIRS

# Lab sessions

- One lab assignment this year, in student pairs — Tuesday mornings (08:45), assistance and feedback on the spot.
- Two parts: algorithm implementation (Distances / Matrices / Retrieval) and a pipeline (transform → analyse → visualise). No Kaggle this year.
- Example: implement DTW & PCA from scratch, then build and evaluate the pipeline.
- Everything is built from scratch — no scikit-learn or other ML toolkits.
- Help channels: Queue, Mattermost, Answers EWI (tag CSE2525), Weblab, Peer. Please do not use e-mail — it won't be answered. Pair up early and register on Brightspace.

LABS 30% · EXAM 70%

# Lab evaluation & grading

HOW THE LAB IS GRADED — 30% OF THE FINAL GRADE

Algorithmic component — evaluated automatically. Pipeline component — peer-reviewed (each submission gets 4 reviews; a penalty if you skip your own reviews — we spot-check quality).

No Kaggle competition this year. No posted solutions — ask for help during the lab sessions.

THE ASSESSMENT

# Final exam

— Final exam = 70%, on WebLab (Osiris + WebLab registration). Open and multiple-choice questions.

— No programming questions this year · closed book, calculator allowed · one resit.

— No partial transfer from previous years. mytimetable.tudelft.nl is authoritative for dates.

WHAT WE ASSUME YOU ALREADY KNOW

# Expected prior knowledge

— Discrete maths — sets, intersections, unions.

— Linear algebra — matrix multiplication, projections, eigenanalysis.

— Probability & statistics — Gaussians, correlation, covariance.

— Graph theory — adjacency matrix, degree, clique, bipartite graph, shortest path.

— Data structures — hash tables and indexes. Programming — Python.

— Machine learning — logistic regression, random forest, SVM, …

Based on CSE1100/1305/1200/1205/1210 and CSE2510/2520. Not enforced — you are responsible for your own study success.

RECAP OF PART I

# Logistics in one slide

KEY POINTS

Questions through online tools and at sessions. One mandatory lab — automated tests + peer review (no Kaggle). Closed-book exam.

This year's focus: information access — retrieval, ranking and evaluation replace anomaly detection and sketching.

Feedback only: dm-cs-ewi@tudelft.nl.

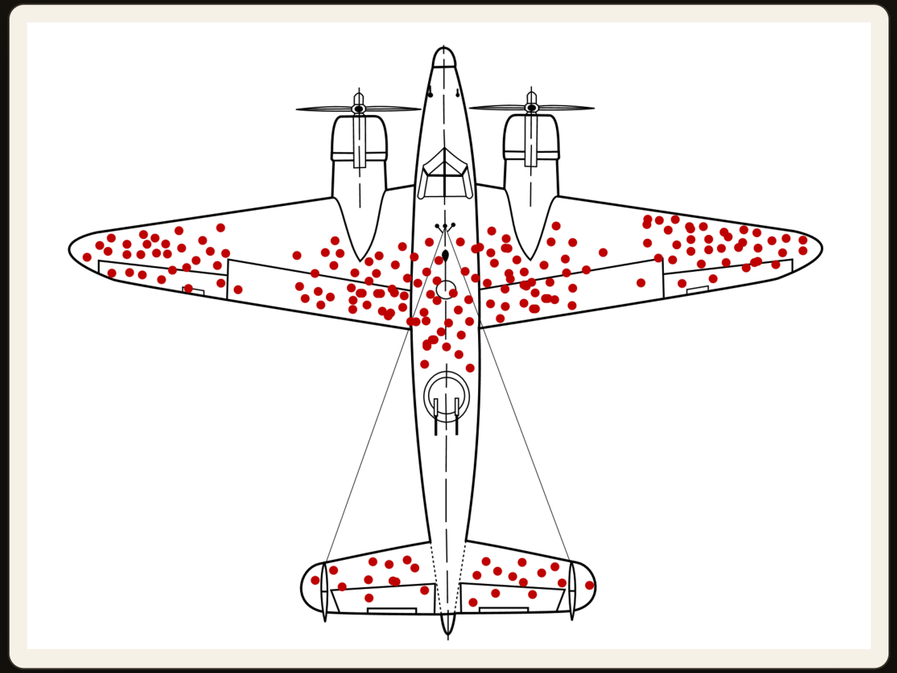
PART III

# A word of caution

*Don’t fool yourself. The rest of this lecture is cautionary tales about results that look great and are wrong.*

WWII · ABRAHAM WALD

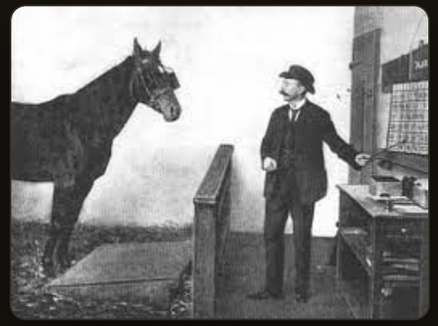
# Survivorship bias



Red dots = damage on planes that returned. Where should you add armour? Not where the dots are — where they aren't (engines, cockpit): planes hit there never came back. Your data is only the survivors.

THE HORSE THAT COULD “COUNT”

# The Clever Hans effect



Hans seemed to do arithmetic — but was really reading his trainer’s unconscious cues. Models do this too: right answer, wrong reason.

TOO GOOD TO BE TRUE

# Shortcuts in learning — Wikipedia citations

THE TASK

Automatically predict which Wikipedia sentences **need a citation**. A neural network scored **100% on the test set** — while humans agreed only **<55%** of the time.

WHAT SHOULD YOU ASK?

100% accuracy on a task humans find genuinely hard is too good to be true — a red flag, not a trophy.

The catch: sentences needing a citation already carried the marker "[ ]". The model learned *"if the bracket is there, predict citation-needed"* — a shortcut, not understanding.

THE CANONICAL ML PARABLE

# Shortcut learning — the “Russian tanks”

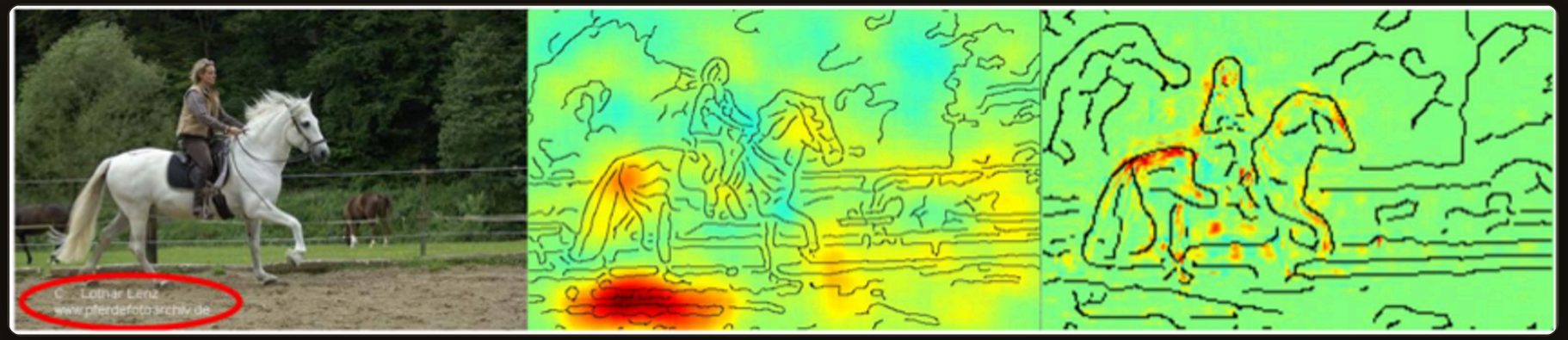




100% accurate in training, worse than random on new data. The network had learned: “if the sky is blue, no tank; otherwise, tank.” It classified the weather, not the tanks.

EXPLAINABLE AI REVEALS THE TRICK

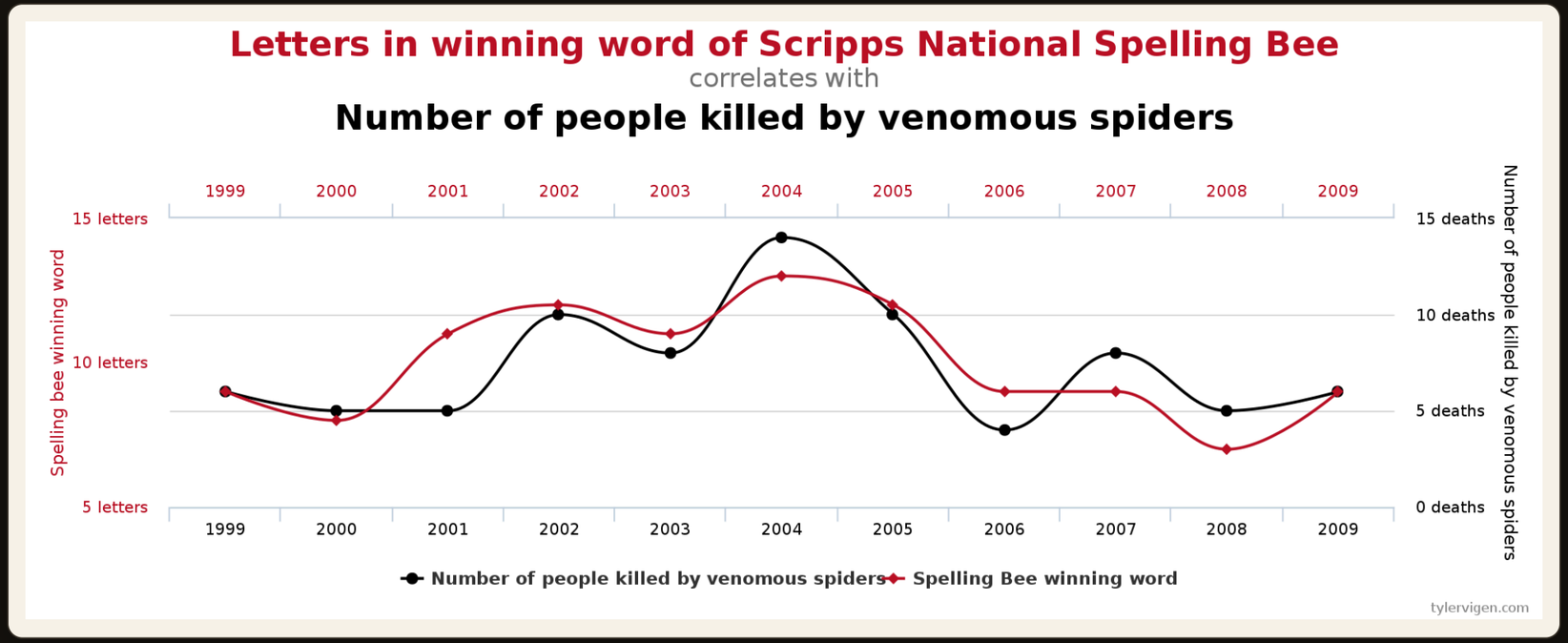
# Shortcuts in image models




A classifier labels this “horse” with high confidence. Saliency maps show *why*: it fires on the copyright watermark in the corner. If “© …” → predict horse.

MULTIPLE COMPARISONS

# You can always find what you're looking for




Test enough hypotheses and some will "correlate" by pure chance. The multiple-comparisons problem is the cause of many data-mining errors — significance you went fishing for isn't significance.

LECTURE 1 · WRAP-UP

# Summary

TAKE AWAY

Data mining = extract useful information from a large volume of data and turn it into an understandable structure. It focuses on the input and output ends of the ML pipeline, often unsupervised.

Organised around three verticals — Distances · Matrices · Retrieval. Grading: labs 30%, exam 70%.

It's called the skill of the 21st century — but tread with caution. Survivorship bias, Clever Hans, shortcut learning and multiple comparisons are waiting to fool you.